Samuele Mattiuzzo

Principal Software Engineer

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Italian / British
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Principal Software Engineer with over 2 decades experience delivering large-scale digital services across UK government and global technology organisations. Experienced in designing cloud-native systems, building scalable APIs, and leading multidisciplinary teams through complex delivery.
Proven track record of improving engineering practices, mentoring developers, and establishing technical standards that enable teams to deliver resilient, user-centred digital services.

TECHNOLOGY & TOOLS

Languages & Frameworks
Python 3 (and 2), Java, Ruby, PHP, JavaScript
Django, Flask, FastAPI, Ruby on Rails, Laravel, Spring Boot, Node.js, React, Cucumber, Selenium

Testing
Postman, SonarQube, Snyk
Automated, exploratory, and regression testing

Cloud & DevOps

AWS (EC2, S3, RDS, DynamoDB, CloudWatch, SNS, IAM), Docker, Kubernetes, Terraform, Helm
GitHub, GitLab, GitHub Actions,, Heroku, DigitalOcean, Splunk

Methodologies
Software architecture design, microservices, monolith modernisation, refactoring,
RESTful, SOAP, GraphQL APIs, Agile (Scrum, Kanban),
TDD, BDD, paired/mob programming
CI/CD pipeline design, security & compliance guardrails

PROFESSIONAL EXPERIENCE

Mastek - Senior Principal Specialist

June 202 - present | Hybrid, Leeds(UK)

Senior technical specialist delivering software engineering solutions for public sector clients, contributing to technical design, engineering quality, and continuous improvement.

Client engagements via Mastek:

NHS Digital - Senior Principal Engineer (*June 2026 – present*):

Tech: Python, FastAPI, PostgreSQL, AWS, Terraform, GitHub CI/CD

Working on the Care Identity Management (CIM) programme, contributing to identity services and reporting capabilities.

- Collaborated with the Delivery Manager and Business Analyst to shape requirements and design and build a prototype for reporting capabilities.
- Optimised automated tests, reducing execution time by 64%.
- Introduced improvements to team ceremonies, particularly retrospectives, to support more effective discussions and continuous improvement.
- Investigate complex issues across application code, databases, AWS infrastructure, and CI/CD pipelines.
- Maintain technical documentation to support knowledge sharing and system understanding.
- Contribute to code reviews, automated testing, and CI/CD improvements to support reliable delivery.

Made Tech - Lead Software Engineer

Apr 2022 - June 2026 | Remote (UK)

Lead engineer delivering digital transformation programmes across multiple UK government departments within blended teams of civil servants, suppliers, and multidisciplinary specialists. Responsible for technical leadership, architecture design, and improving delivery capability across teams.

Promoted from Senior Software Engineer to Lead Software Engineer based on performance and delivery across complex government services.

Actively contribute to Made Tech's internal engineering and organisational leadership initiatives.

- **People Forum Representative**: member of the internal forum representing employees in discussions with senior leadership. Contribute to shaping company policies, benefits and organisational initiatives, acting as a bridge between employees and executive leadership.
- **Community of Practice Lead**: organise and facilitate internal engineering communities to share knowledge, promote best practices, and strengthen cross-team technical collaboration.
- **Mentorship Programme Mentor**: mentor engineers across the organisation, supporting career development, technical growth, and progression within the company.

Client engagements via Made Tech:

Ministry of Justice - Lead Software Engineer (*Sept 2025 – June 2026*):

Tech: Java, Spring Boot, PostgreSQL, AWS, Terraform, Helm, GitHub CI/CD

Working within the Data Stewardship team designing and delivering a new API platform and central datastore enabling secure cross-service data access.

- Leading architecture and development of a scalable API platform supporting data sharing across services.
- Establishing architectural decision records (ADRs) and engineering guardrails to guide long-term platform evolution.
- Defining patterns for secure service integration, data storage, and access control.
- Collaborating with platform engineering, security, and delivery teams across the organisation.

Ministry of Housing, Communities and Local Government - Lead Software Engineer (*Jan 2025 – Sept 2025*):

Tech: Python, Django, PostgreSQL, Docker, AWS, Terraform

Technical lead responsible for engineering delivery across two development teams and one infrastructure team.

- Introduced delivery practices including Three Amigos and structured backlog refinement.
- Implemented CI/CD pipelines and automated release processes enabling iterative delivery to stakeholders.
- Improved delivery performance, doubling team velocity within three sprints.
- Worked closely with UX, product, and data teams to remove uncertainty and unblock development.
- Fostered a culture of ownership, accountability, and psychological safety within the teams.

Department for Business and Trade - Lead Software Engineer (*Oct 2025 – Dec 2024*):

Tech: Python, Django, React, Docker, AWS, PostgreSQL

Led development of an internal financial platform supporting budget management and operational oversight.

- Delivered a cloud-native system improving financial transparency and operational oversight.
- Supported SRE teams in migrating infrastructure from an internal cloud platform to AWS.
- Improved scalability and reliability through infrastructure modernisation.
- Mentored engineers and strengthened engineering practices across the team.

Department for Levelling Up, Housing & Communities - Senior Software Engineer (*Jul 2022 – Sept 2023*):

Tech: Python, Django, Ruby on Rails, Docker, AWS

Team lead responsible for the build of the Ukraine Resettlement digital service.

- Delivered a production-ready service from prototype in just over six weeks.
- Led technical design of a secure system managing sensitive government data.
- Worked closely with user researchers and stakeholders to ensure the service met operational needs.
- Guided engineering delivery under significant time pressure.

Simpleweb - Senior Software Engineer

Nov 2020 – Feb 2022 | Bristol, United Kingdom

Delivered multiple client projects across a range of industries.

- Led architecture and delivery of short-cycle product builds.
- Worked directly with clients to shape technical solutions aligned with business needs.
- Managed development, deployment, and operational processes within a small engineering team.
- ***Tech****: Ruby on Rails, PHP (Laravel), Postgres, MongoDB, Node.js, Angular JS, AWS (S3), Docker, Heroku*

Athonet - Senior Python Engineer

Aug 2020 - Nov 2020 | Bolzano Vicentino, Vicenza, Italy

Worked within the R&D division building backend systems supporting 5G telecom infrastructure.

- Developed scalable backend services supporting next-generation telecom network features.
- Collaborated with distributed engineering teams focused on performance and reliability.
- ***Tech stack****: Python, Django, Postgres, Docker, Elixir*

Anthesis Group - Senior Software Engineer

Jan 2020 - Jul 2020 | Bristol, England, United Kingdom

Technical lead on a government-backed platform tracking waste movements and regulatory compliance.

- Designed and implemented backend architecture supporting environmental data workflows.
- Delivered infrastructure and deployment pipelines for the platform.
- Presented technical progress to stakeholders and ensured compliance requirements were met.
- ***Tech stack****: Python, Django, MySQL, Docker, Fabric, GovUK Design System.*

Development Initiatives - Senior Software Engineer

Sept 2018 - Dec 2019 | Bristol, England, United Kingdom

Senior engineer and technical community lead for the International Aid Transparency Initiative (IATI)

- Maintained and evolved core tools supporting international open data standards.
- Refactored and consolidated repositories to reduce technical overhead and improve maintainability.
- Introduced a community code of conduct supporting collaboration across international con
- ***Tech stack****: Python, Django, Flask, Docker, JavaScript, CSS, XML, open data standards*

Potato (now AKQA Leap) - Python/Django Developer

Feb 2013 - Aug 2018 | London & Bristol, England, United Kingdom

Worked on several projects for Google.

- Contributed to backend systems for the Google I/O ticketing platform.
- Developed scalable backend services capable of supporting high user demand.
- Delivered interactive multimedia experiences including an award-winning 3D project.
- ***Tech stack****: Python, Django, Google Appengine, Javascript, Node.js, Angular JS, CSS, HTML.*

SIAV S.P.A. - Python/Django Developer

Jun 2011 - Feb 2013 | Milano, Italy

Full-stack developer working on enterprise web applications.

- Implemented backend systems and advanced search functionality using Apache Solr.
- Delivered frontend interfaces using JavaScript and modern web technologies.
- ***Tech stack:*** *Python, Django, Apache-Solr, Javascript, JQuery, CSS, HTML*

Kasato S.R.L. - Python/Django Developer

Jan 2009 - Jun 011 | Padova, Italy

Developer within a small startup building digital services for funeral providers.

- Developed a social platform connecting families with service providers.
- Built a digital obituary display device with adaptive brightness based on ambient light.
- ***Tech stack****: Python, Django, Javascript, CSS, HTML, C++*

EDUCATION

Università Ca' Foscari di Venezia, Italy (2009 - 2011) – Computer Science
Università degli Studi di Padova, Italy (2005 - 2009) – Computer Science

CERTIFICATIONS/QUALIFICATIONS

AWS Certified Cloud Practitioner: Amazon Web Services
Palantir Foundry Foundations: Palantir Technologies
IELTS C1 English
Security Cleared